

Teorema de Cauchy-Goursat para rectángulos con singularidades

Teorema 1 (de Cauchy-Goursat para rectángulos con singularidades).

Sea Ω un dominio, $R \subset \Omega$ un rectángulo y sean $\{z_k\}_{k=1}^N \in \overset{\circ}{R}$ tales que $f \in \mathcal{H}(\Omega \setminus \{z_k\}_{k=1}^N)$

$$\forall k \in \mathbb{N}_N : \lim_{z \rightarrow z_k} (z - z_k) f(z) = 0 \implies \int_{\partial R} f(z) dz = 0.$$

Demostración: Lo vamos a demostrar solamente para $N = 1$. Vamos a aislar z_1 en un cuadrado R_0 de tal forma que R queda dividido en 9 rectángulos (uno de ellos será R_0).

Sabemos que $\lim_{z \rightarrow z_1} (z - z_1) f(z) = 0$, i.e. $\forall \varepsilon > 0 : \exists \delta > 0 : |z - z_1| < \delta \implies |f(z)| < \frac{\varepsilon}{|z - z_1|}$.

Tomaremos $R_0 \subset D(z_1, \delta)$.

$$\int_{\partial R} f(z) dz = \sum_{j=1}^8 \int_{\partial R_j} f(z) dz + \int_{\partial R_0} f(z) dz.$$

Por el Teorema de Cauchy-Goursat para rectángulos, $\sum_{j=1}^8 \int_{\partial R_j} f(z) dz = 0$.

$$\implies \left| \int_{\partial R} f(z) dz \right| = \left| \int_{\partial R_0} f(z) dz \right| \leq \int_{\partial R_0} |f(z)| |dz| \leq \varepsilon \int_{\partial R_0} \frac{1}{|z - z_1|} |dz|.$$

Entonces, $\forall \varepsilon > 0 : \left| \int_{\partial R} f(z) dz \right| \leq \frac{2\varepsilon}{l_0} \int_{\partial R_0} |dz| = \frac{2\varepsilon}{l_0} \cdot \underbrace{\text{long}(\partial R_0)}_{4l_0} = 8\varepsilon \implies \int_{\partial R} f = 0. \blacksquare$

Temporary page!

L^AT_EX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because L^AT_EX now knows how many pages to expect for this document.